6-6 Numerical Calculations [2]

Point!

1 Representation of Negative Numbers Using Complements

([1]**Complement**) : The smallest number which, when added to a given natural number, produces a carry to the next higher digit. Complements are used for representing negative numbers in computers.

[1] ([2]**10's complement**) : In decimal, the smallest number which, when added, produces a carry to the next higher digit.

<Examples> The 10's complement of 71 is ([3]**29**), and the 10's complement of 635 is ([4]**365**).

[2] ([5]**2's complement**) : In binary, the smallest number which, when added, produces a carry to the next higher digit.

<How to calculate the 2's complement>

For each digit of the original binary number, invert 0s to 1s and 1s to 0s, then add 1 at the end to mechanically obtain the result.

<Example> The 2's complement of $0101_{(2)}$

By inverting each digit from 0 to 1 and from 1 to 0, it becomes ([6]**$1010_{(2)}$**). Finally, by adding 1, it becomes ([7]**$1011_{(2)}$**).

2 Subtraction Using Complements

A computer uses subtraction by performing addition with complements.

<Subtraction process>

1) Find the ([8]**complement**) of the subtrahend.

2) Use the complement to perform ([9]**addition**).

3) Ignore the leading digits of the calculation result and provide the answer.

<Example 1>

Subtraction of "8 − 6" in decimal

1) The 10's complement of the decimal number 6 is ([10]**4**)

2) 8 − 6 ⇒ 8 ([11]+ 4)
= ([12]**12**)

3) Ignoring the leading digit, ([13]2)

<Example 2>

Subtraction of "$1000_{(2)}$ − $0111_{(2)}$" in binary

1) The 2's complement of the binary number $0111_{(2)}$ is ([14]$1001_{(2)}$)

2) $1000_{(2)}$ − $0111_{(2)}$ ⇒ $1000_{(2)}$ ([15]+ $1001_{(2)}$)
= ([16]$10001_{(2)}$)

3) Ignoring the leading digit, ([17]$0001_{(2)}$)

Warm Up

Answer the following questions.

(1) What is the complement of the following binary number?

[1] $1001_{(2)}$ [2] $0100_{(2)}$

(2) Use complements to perform the following binary subtractions.

[1] $1100_{(2)} - 0111_{(2)}$ [2] $1110_{(2)} - 1001_{(2)}$

Explanation

(1) [1] By inverting each digit from 0 to 1 and from 1 to 0, it becomes $0110_{(2)}$.
Finally, by adding 1, it becomes $\underline{0111}_{(2)}$.

[2] By inverting each digit from 0 to 1 and from 1 to 0, it becomes $1011_{(2)}$.
Finally, by adding 1, it becomes $\underline{1100}_{(2)}$. Adding 1 causes a place value to carry over.

(2) [1] The complement of the binary number $0111_{(2)}$ is $1001_{(2)}$, so
$1100_{(2)} - 0111_{(2)} \Rightarrow 1100_{(2)} + 1001_{(2)} = 10101_{(2)}$
By ignoring the leading digit, $\underline{0101}_{(2)}$

[2] The complement of the binary number $1001_{(2)}$ is $0111_{(2)}$, therefore,
$1110_{(2)} - 1001_{(2)} \Rightarrow 1110_{(2)} + 0111_{(2)} = 10101_{(2)}$
By ignoring the leading digit, $\underline{0101}_{(2)}$

Try

Answer the following questions.

(1) What is the complement of the following binary number?

[1] $0101_{(2)}$ [2] $1101_{(2)}$ [3] $10110001_{(2)}$ [4] $01001100_{(2)}$

(2) Use complements to perform the following binary subtractions.

[1] $1101_{(2)} - 0110_{(2)}$ [2] $1010_{(2)} - 0111_{(2)}$ [3] $1101_{(2)} - 1010_{(2)}$

Exercise

1 Answer the following questions.

(1) What is the complement of the following binary number?

[1] $0111_{(2)}$ [2] $1011_{(2)}$ [3] $10011011_{(2)}$ [4] $11000110_{(2)}$

(2) Use complements to perform the following binary subtractions.

[1] $1010_{(2)} - 0110_{(2)}$ [2] $1011_{(2)} - 1001_{(2)}$ [3] $1100_{(2)} - 0010_{(2)}$